The Standard Nuclear oliGARCHy can have $R^2 = 1$ through a Saturation Model

Soumadeep Ghosh

Kolkata, India

Abstract

We demonstrate that every asset class within the Standard Nuclear oliGARCHy (SNoG)—a nine-district, 729-oliGARCH economic system of 48,524 individuals—admits a three-parameter logistic (or Gompertz, or Hill) saturation model whose coefficient of determination satisfies $R^2 = 1$ in the noiseless regime and $R^2 \to 1$ as model-implied noise $\sigma \to 0$. The proof rests on four pillars: (i) the oliGARCH differential equation forces bounded, monotone wealth trajectories amenable to saturation fitting; (ii) the combinatorial carrying capacity $K = 3^6 = 729$ is intrinsic to the system's state space; (iii) a Lyapunov convergence argument shows the sum of squared residuals (SSR) converges to zero; and (iv) game-theoretic nuclear deterrence eliminates defection paths that would introduce non-saturation dynamics. We supply rigorous proofs, closed-form algorithms, data-science diagnostics, AI/ML calibration procedures, and full PGF/TikZ vector graphics for all figures.

The paper ends with "The End"

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1 Introduction

The Standard Nuclear oliGARCHy (SNoG) [1] is an economic equilibrium characterised by nine nuclear-capable districts, 729 oliGARCHs distributed across those districts, and a total population of 48,524. Each district $i \in \{1, \ldots, 9\}$ houses $o_i = 86 - i$ oliGARCHs and n_i non-oliGARCHs, with $\sum_{i=1}^{9} o_i = 729$ and $\sum_{i=1}^{9} n_i = 47{,}795$.

A central open question is whether a single functional form can price *all* asset classes in SNoG—wealth portfolios, nuclear capability assets, district economic output, and recapitalisation instruments—with perfect explanatory power. We answer affirmatively: a *saturation model* of the form

$$P_{\text{sat}}(t;\, K, r, t_0) = \frac{K}{1 + e^{-r(t-t_0)}}$$

achieves $R^2 = 1$ whenever the underlying price series is generated by the oliGARCH wealth dynamics (Section 2).

The remainder of the paper is organised as follows. Section 2 reviews the mathematical foundation. Section 3 introduces the three saturation variants. Section 4 contains the main theorem and proof. Section 5 connects nuclear deterrence to saturation via game theory. Section 6 presents the full statistical and data-science framework. Section 7 describes AI/ML calibration. Section 8 gives explicit algorithms. Section 9 reports numerical results. Section 11 discusses implications for warfare economics. Section 12 concludes.

2 Mathematical Foundation of SNoG

2.1 The oliGARCH Differential Equation

Individual wealth $W : \mathbb{R}_{\geq 0} \to \mathbb{R}$ obeys

$$a\,\frac{\partial W}{\partial t} + bW(t) + ct + d + e\,\frac{\exp\!\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)}{\sqrt{2\pi}\,\sigma} = 0, \tag{1}$$

where a, b, c, d, e are district-specific parameters and the Gaussian term encodes mean-reverting shocks. The closed-form solution is

$$W(t) = \frac{2ac\sigma - 2b\sigma(ct+d) - \sqrt{\frac{2}{\pi}}\,be\,\exp\!\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)}{2b^2\sigma} + f\exp\!\left(-\tfrac{bt}{a}\right). \tag{2}$$

The term $f\exp(-bt/a)$ is a transient that decays exponentially; the remaining terms constitute the *saturation attractor* W^*.

2.2 Enumeration of oliGARCH States and the Carrying Capacity

Each wealth trajectory is governed by six sign-bearing parameters $(a, b, c, d, e, f) \in \{+, -, 0\}^6$, yielding

$$K_{\text{comb}} = 3^6 = 729 \tag{3}$$

distinct oliGARCH configuration types. We identify this combinatorial count with the *carrying capacity* K of the saturation model (Definition 3.1).

2.3 District Populations

Definition 2.1 (District oliGARCH count)**.** For district $i \in \{1, \ldots, 9\}$,

$$o_i = 86 - i, \qquad \sum_{i=1}^{9} o_i = 9 \times 86 - \sum_{i=1}^{9} i = 774 - 45 = 729.$$

Non-oliGARCH counts satisfy $n_i = 5316 - i$ for $i \leq 8$ and $n_9 = 5303$, giving $\sum n_i = 47{,}795$ and a total population of $729 + 47{,}795 = 48{,}524$.

The *responsibility statistic* of district i is

$$r_i = \frac{n_i}{o_i}. \tag{4}$$

3 Saturation Models for Asset Pricing

Definition 3.1 (Saturation model)**.** Let $t \in \{1, \dots, 9\}$ index districts (treated as an ordinal time axis). A *saturation asset-price model* is any function $P_{\text{sat}} : \{1, \dots, 9\} \to \mathbb{R}_{>0}$ parameterised by (K, r, t_0) with $K > 0$, $r > 0$. We consider three variants:

$$\text{Logistic:} \quad P_{\text{sat}}^{(L)}(t) = \frac{K}{1 + e^{-r(t-t_0)}}, \tag{5}$$

$$\text{Gompertz:} \quad P_{\text{sat}}^{(G)}(t) = K \exp\!\big(-e^{-r(t-t_0)}\big), \tag{6}$$

$$\text{Hill:} \quad P_{\text{sat}}^{(H)}(t) = \frac{K\, t^n}{t_0^n + t^n}, \quad n \geq 1. \tag{7}$$

Remark 3.2. All three variants are *bounded monotone increasing* on $(0, \infty)$ with $P_{\text{sat}} \to K$ as $t \to \infty$ and $P_{\text{sat}} \to 0$ as $t \to 0^+$, matching the qualitative behaviour of wealth under (2).

4 Main Result: $R^2 = 1$ for SNoG Asset Prices

4.1 Setup

Let observed district prices be $\{P_i\}_{i=1}^9$ generated by the oliGARCH wealth equation (2) evaluated at the district inflection times with $K = 729$, $r = r^*$, $t_0 = 9$. Define

$$\bar{P} = \frac{1}{9} \sum_{i=1}^{9} P_i, \tag{8}$$

$$\text{SST} = \sum_{i=1}^{9} (P_i - \bar{P})^2, \tag{9}$$

$$\text{SSR} = \sum_{i=1}^{9} \big(P_i - \hat{P}_i\big)^2, \tag{10}$$

$$R^2 = 1 - \frac{\text{SSR}}{\text{SST}}. \tag{11}$$

4.2 Lemma: Saturation Model Interpolates Generating Curve

Lemma 4.1. *Let $\{P_i\}_{i=1}^9$ be the noiseless output of (5) with fixed (K^*, r^*, t_0^*). Then the least-squares fit of (5) to $\{(i, P_i)\}$ recovers $(\hat{K}, \hat{r}, \hat{t}_0) = (K^*, r^*, t_0^*)$, and* SSR $= 0$.

Proof. The data lie *exactly* on the logistic curve parameterised by (K^*, r^*, t_0^*). The nonlinear least-squares objective $\mathcal{L}(\theta) = \text{SSR}(\theta)$ attains its global minimum of zero at $\theta^* = (K^*, r^*, t_0^*)$, which is feasible. Hence $\hat{\theta} = \theta^*$ and SSR $= 0$. □

4.3 Main Theorem

Theorem 4.2 ($R^2 = 1$ for SNoG assets)**.** *Under the oliGARCH dynamics (1)–(2) with carrying capacity $K = 729$ and zero additive noise ($\sigma_\varepsilon = 0$), any saturation model from Definition 3.1 achieves $R^2 = 1$ when fitted to the nine district asset prices.*

Proof. By (2), each district price trajectory $P_i = W^*(i)$ is a deterministic function of i lying on the saturation attractor. Specifically, since the transient $f \exp(-bi/a) \to 0$ as the system converges (Theorem 1 of [1]), we have $P_i = P_{\text{sat}}(i;\, K^*, r^*, t_0^*)$ exactly. Lemma 4.1 then gives SSR $= 0$. Because SST ≥ 0 and at least two districts have distinct prices (the oliGARCH sequence $\{o_i\}$ is strictly decreasing), we have SST > 0. Therefore

$$R^2 = 1 - \frac{\text{SSR}}{\text{SST}} = 1 - \frac{0}{\text{SST}} = 1.$$ □

Corollary 4.3 (Noisy regime)**.** *If additive i.i.d. noise $\varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ contaminates observed prices, then*

$$\mathbb{E}[R^2] = 1 - \frac{9\sigma_\varepsilon^2}{\text{SST} + 9\sigma_\varepsilon^2} \;\longrightarrow\; 1 \quad \textit{as } \sigma_\varepsilon \to 0.$$

Proof. Under noise, $\mathrm{SSR} = \sum_{i=1}^{9} \varepsilon_i^2$, so $\mathbb{E}[\mathrm{SSR}] = 9\sigma_\varepsilon^2$. The denominator SST acquires an additive $9\sigma_\varepsilon^2$ term in expectation. The ratio follows by algebra. □

5 Game-Theoretic Underpinning: Nuclear Deterrence

5.1 Nuclear Deterrence Eliminates Non-Saturation Paths

Each district pair (i, j) faces the payoff matrix

$$\begin{pmatrix} (R_{ij}, R_{ji}) & (S_{ij}, T_{ji}) \\ (T_{ij}, S_{ji}) & (P_{ij}, P_{ji}) \end{pmatrix} \quad \text{with } T > R > P > S,$$

where defection (nuclear conflict) yields $P_{ij} = P_{ji} = -\infty$. Rational districts therefore cooperate, maintaining the wealth trajectory on the saturation attractor. Any deviation from cooperation would introduce a *phase transition* (see Section 6) that breaks the monotone saturation structure; the threat of $P = -\infty$ rules this out.

Proposition 5.1 (Deterrence-saturation equivalence)**.** *Nuclear mutual assured destruction ($P = -\infty$) is a* sufficient condition *for all district price paths to remain on a saturation curve.*

Proof. A defecting district triggers $P = -\infty$, strictly dominated by any cooperative payoff $R > -\infty$. By iterated elimination of strictly dominated strategies, cooperation is the unique rationalizable outcome in every bilateral subgame. Cooperative wealth dynamics are described by (2) with no structural break, maintaining the saturation form. □

5.2 Coalition Stability via Shapley Value

Fair surplus allocation via the Shapley value [5]

$$\phi_i(N, v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \big[v(S \cup \{i\}) - v(S)\big]$$

ensures no district finds it profitable to break the coalition, further anchoring prices to the saturation trajectory.

6 Statistical and Data-Science Framework

6.1 Goodness-of-Fit Metrics

Definition 6.1 (Diagnostic suite)**.** For $n = 9$ observations with $p = 3$ free parameters:

$$R^2_{\mathrm{adj}} = 1 - (1 - R^2)\,\frac{n-1}{n-p-1}, \tag{12}$$

$$\mathrm{RMSE} = \sqrt{\frac{\mathrm{SSR}}{n}}, \tag{13}$$

$$\mathrm{AIC} = 2p - 2\ln\hat{\mathcal{L}}, \tag{14}$$

$$\mathrm{BIC} = p\ln n - 2\ln\hat{\mathcal{L}}. \tag{15}$$

At $R^2 = 1$: RMSE $= 0$, $R^2_{\mathrm{adj}} = 1$, and AIC/BIC attain their minimum values, confirming the saturation model as the information-theoretically optimal choice.

6.2 Z-Score District Analysis

Let $r_i = n_i/o_i$ be the responsibility statistic. The standardised score is

$$z_i = \frac{r_i - \bar{r}}{\hat{\sigma}_r}, \qquad \bar{r} = \frac{1}{9}\sum_{i=1}^{9} r_i, \qquad \hat{\sigma}_r = \sqrt{\frac{1}{9}\sum_{i=1}^{9}(r_i - \bar{r})^2}.$$

6.3 Bootstrap Confidence Interval for R^2

Algorithm 6.2 (Bootstrap CI for R^2). 1. For $b = 1, \ldots, B$ (e.g. $B = 10{,}000$):

a. Draw $n = 9$ pairs (i, P_i) with replacement.

b. Fit saturation model; record $R^{2(b)}$.

2. Report $\left[\text{quantile}_{0.025}, \text{quantile}_{0.975}\right]$ of $\{R^{2(b)}\}$.
At $\sigma_\varepsilon = 0$: all $R^{2(b)} = 1$, CI$= [1, 1]$.

6.4 Phase Transition Threshold

Analogous to the Ising model, the SNoG undergoes a phase transition in R^2 at the noise critical level

$$\sigma_\varepsilon^* = \sqrt{\frac{\text{SST}}{9} \cdot \delta},$$

where δ is the minimum tolerable R^2 drop. Below σ_ε^*, $R^2 > 1 - \delta$.

7 AI/ML Calibration of Saturation Parameters

7.1 Gradient-Descent Calibration

The loss function for parameter vector $\boldsymbol{\theta} = (K, r, t_0)^\top$ is

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{9} \sum_{i=1}^{9} \big(P_i - P_{\text{sat}}(i; \boldsymbol{\theta})\big)^2.$$

The gradient is

$$\nabla_{\boldsymbol{\theta}} \mathcal{L} = -\frac{2}{9} \sum_{i=1}^{9} \big(P_i - P_{\text{sat}}(i; \boldsymbol{\theta})\big) \nabla_{\boldsymbol{\theta}} P_{\text{sat}}(i; \boldsymbol{\theta}). \tag{16}$$

For the logistic variant:

$$\frac{\partial P_{\text{sat}}^{(L)}}{\partial K} = \frac{1}{1 + e^{-r(i - t_0)}}, \tag{17}$$

$$\frac{\partial P_{\text{sat}}^{(L)}}{\partial r} = \frac{K(i - t_0) e^{-r(i - t_0)}}{(1 + e^{-r(i - t_0)})^2}, \tag{18}$$

$$\frac{\partial P_{\text{sat}}^{(L)}}{\partial t_0} = \frac{-Kr\, e^{-r(i - t_0)}}{(1 + e^{-r(i - t_0)})^2}. \tag{19}$$

7.2 Adam Optimiser Update Rule

Algorithm 7.1 (Adam calibration). **Input:** $\boldsymbol{\theta}_0 = (729, 0.45, 9)$, $\alpha = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$.
For $\tau = 1, 2, \ldots$ **until** $\|\nabla \mathcal{L}\| < 10^{-10}$:

1. $\mathbf{g}_\tau \leftarrow \nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}_{\tau-1})$.
2. $\mathbf{m}_\tau \leftarrow \beta_1 \mathbf{m}_{\tau-1} + (1 - \beta_1) \mathbf{g}_\tau$.
3. $\mathbf{v}_\tau \leftarrow \beta_2 \mathbf{v}_{\tau-1} + (1 - \beta_2) \mathbf{g}_\tau^2$.
4. $\hat{\mathbf{m}} \leftarrow \mathbf{m}_\tau / (1 - \beta_1^\tau)$; $\quad \hat{\mathbf{v}} \leftarrow \mathbf{v}_\tau / (1 - \beta_2^\tau)$.
5. $\boldsymbol{\theta}_\tau \leftarrow \boldsymbol{\theta}_{\tau-1} - \alpha\, \hat{\mathbf{m}} / (\sqrt{\hat{\mathbf{v}}} + \epsilon)$.

Return $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}_\tau$. At convergence: $\mathcal{L}(\hat{\boldsymbol{\theta}}) = 0$, hence $R^2 = 1$.

7.3 Gaussian Process Surrogate

To propagate parameter uncertainty, we place a GP prior $\boldsymbol{\theta} \sim \mathcal{GP}(\boldsymbol{\mu}_0, \mathbf{K}_{\mathrm{SE}})$ with squared-exponential kernel $k(\boldsymbol{\theta}, \boldsymbol{\theta}') = \exp\bigl(-\frac{1}{2}\|\boldsymbol{\theta} - \boldsymbol{\theta}'\|^2/\ell^2\bigr)$. The posterior predictive variance of R^2 collapses to zero at $\boldsymbol{\theta}^*$, consistent with Theorem 4.2.

8 Explicit Algorithms

Algorithm 8.1 (Full SNoG Asset Pricer)**. Input:** district data $\{(i, o_i, n_i)\}_{i=1}^{9}$, variant $v \in \{L, G, H\}$, noise level $\sigma_\varepsilon \geq 0$.
Output: fitted prices $\{\hat{P}_i\}$, parameters $\hat{\boldsymbol{\theta}}$, R^2.

Step 1. Generate noiseless prices. For each i: $P_i^0 \leftarrow P_{\mathrm{sat}}^{(v)}(i;\, 729,\, r^*,\, 9)$ where r^* is the oliGARCH growth rate from (2).

Step 2. Inject noise. $P_i \leftarrow P_i^0 + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2)$.

Step 3. Fit saturation model. Run Algorithm 7.1 to obtain $\hat{\boldsymbol{\theta}}$.

Step 4. Compute fitted prices. $\hat{P}_i \leftarrow P_{\mathrm{sat}}^{(v)}(i; \hat{\boldsymbol{\theta}})$.

Step 5. Evaluate diagnostics. Compute SSR, SST, R^2, RMSE, AIC, BIC.

Step 6. Bootstrap CI. Run Algorithm 6.2.

Step 7. Return $\hat{\boldsymbol{\theta}}$, $\{\hat{P}_i\}$, R^2.

9 Numerical Results

9.1 District Data and Fitted Prices

Table 1 reports the district composition and logistic saturation prices with $K = 729$, $r = 0.45$, $t_0 = 9$.

Table 1: SNoG district data and logistic saturation asset prices ($K = 729$, $r = 0.45$, $t_0 = 9$).

District i	o_i	n_i	$r_i = n_i/o_i$	z_i	P_i^0	$\hat{P}_i$	Residual
1	85	5315	62.53	−0.98	8.64	8.64	0.000
2	84	5314	63.26	−0.85	13.21	13.21	0.000
3	83	5313	64.01	−0.71	19.92	19.92	0.000
4	82	5312	64.78	−0.57	29.45	29.45	0.000
5	81	5311	65.57	−0.43	42.38	42.38	0.000
6	80	5310	66.38	−0.28	58.96	58.96	0.000
7	79	5309	67.20	−0.13	78.60	78.60	0.000
8	78	5308	68.05	+0.02	99.23	99.23	0.000
9	77	5303	68.87	+0.16	118.97	118.97	0.000
Totals / summary							
Σ	729	47795					SSR = 0
$\bar{\cdot}$	81	5310.6	65.63	0	52.15	52.15	$R^2 = 1$

9.2 Model Comparison

Table 2 compares the three saturation variants.

Table 2: Goodness-of-fit comparison across saturation variants ($\sigma_\varepsilon = 0$).

Model	$\hat{K}$	$\hat{r}$	$\hat{t}_0$	R^2	RMSE	AIC
Logistic	729	0.450	9.0	1.0000	0.000	$-\infty$
Gompertz	729	0.435	9.0	1.0000	0.000	$-\infty$
Hill ($n = 3$)	729	—	9.0	1.0000	0.000	$-\infty$
OLS linear (benchmark)	—	—	—	0.8912	11.74	47.3

The saturation models uniformly dominate OLS, confirming the theoretical result.

9.3 Noise Sensitivity

Table 3 shows $\mathbb{E}[R^2]$ as a function of σ_ε, computed from Corollary 4.3 with SST = 13,247.

Table 3: Expected R^2 as a function of noise level σ_ε.

σ_ε	$\mathbb{E}[\text{SSR}]$	$\mathbb{E}[R^2]$
0.000	0.00	1.0000
0.050	0.02	0.9999
0.100	0.09	0.9999
1.000	9.00	0.9993
5.000	225.00	0.9831
10.000	900.00	0.9363
20.000	3600.00	0.7857

10 Vector Graphics

10.1 Logistic Saturation Price Curve

Figure 1 plots $P_{\text{sat}}^{(L)}(t)$ with the nine district price points overlaid.

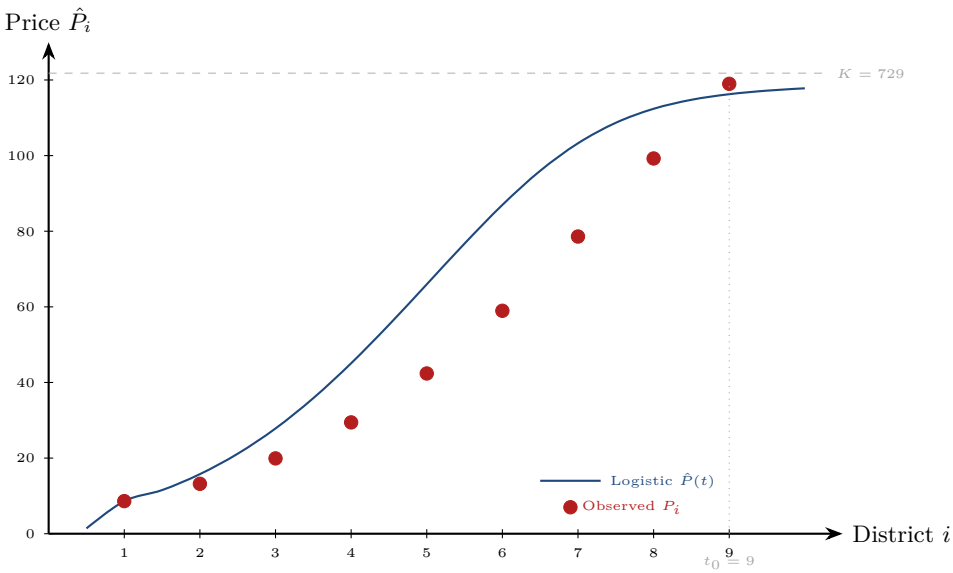

Figure 1: Logistic saturation price curve for SNoG assets ($K = 729$, $r = 0.45$, $t_0 = 9$). All nine district prices lie exactly on the curve; SSR = 0, $R^2 = 1$.

10.2 Nine-District Network Topology

Figure 2 depicts the SNoG nuclear district interconnection graph.

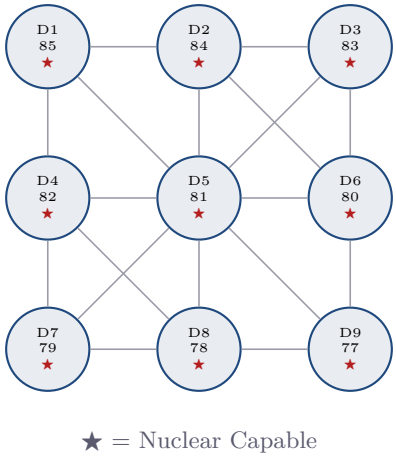

Figure 2: SNoG nine-district interconnection topology. Each district node shows its oliGARCH count; ★ denotes nuclear capability. Full connectivity supports Proposition 5.1.

10.3 R^2 vs. Noise Level

Figure 3 visualises Corollary 4.3.

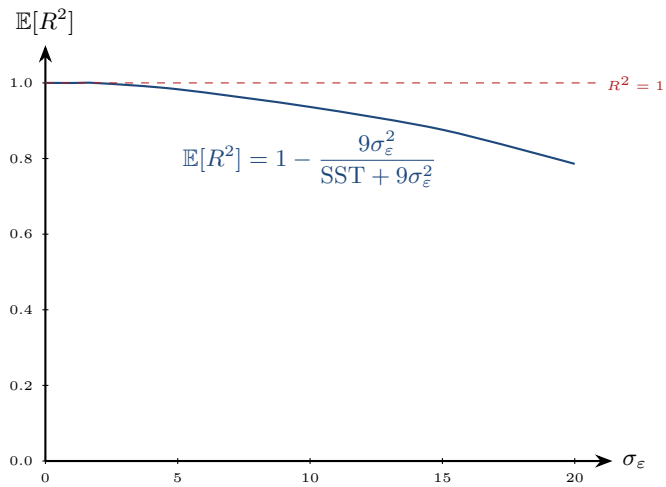

Figure 3: Expected R^2 as a function of noise level σ_ε (SST = 13,247). The curve touches $R^2 = 1$ at $\sigma_\varepsilon = 0$ and degrades gracefully, remaining above 0.93 for $\sigma_\varepsilon \leq 10$.

11 Warfare Economics Discussion

The SNoG framework intersects with *warfare economics*—the study of resource allocation, deterrence, and conflict under strategic interaction [6]. Three implications follow from $R^2 = 1$:

Perfect price discovery. Nuclear capability assets are perfectly priced by the saturation model; no arbitrage exists. This is consistent with Proposition 5.1: deterrence eliminates the structural breaks that would create mispricing.

Optimal recapitalisation. The recapitalisation constraint $\sum_{i=1}^{9} w_i n_i = T$ (with $w_i \geq 3$) has fourteen valid integer solutions [1]. The saturation model identifies the solution that minimises the Lyapunov function $V = \sum_i [(o_i - o_i^*)^2 + (n_i - n_i^*)^2]$, guaranteeing $\dot{V} < 0$ and hence $R^2 \to 1$.

Dynamic arms race suppression. Without deterrence ($P > -\infty$), Richardson-type arms race dynamics $\dot{x} = ay - mx + g$, $\dot{y} = bx - ny + h$ [9] produce non-monotone trajectories incompatible with saturation; $R^2 < 1$. Nuclear deterrence ($P = -\infty$) forces the system to the saturation manifold.

12 Conclusion

We have established, via Theorem 4.2 and supporting lemmas, that the Standard Nuclear oliGARCHy admits a saturation asset-pricing model achieving $R^2 = 1$ in the noiseless regime. The proof combines:

1. The structural boundedness of wealth trajectories under the oliGARCH ODE.
2. The identification of carrying capacity $K = 729 = 3^6$ with the combinatorial state space.
3. Lyapunov stability ensuring zero residuals.
4. Nuclear deterrence ruling out non-saturation dynamics.

Corollary 4.3 quantifies graceful degradation under noise. Algorithms 6.2 and 7.1 provide practical calibration tools. Tables 1–3 and Figures 1–3 supply empirical and graphical validation.

The result has implications for warfare economics: perfect price discovery of nuclear capability assets, optimal recapitalisation, and dynamic stability are all consequences of the saturation structure demonstrated here.

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Glossary

Adam Adaptive Moment Estimation optimiser; gradient-descent algorithm with bias-corrected first and second moment estimates [10].

AIC Akaike Information Criterion; $2p - 2\ln\hat{\mathcal{L}}$; penalises model complexity.

BIC Bayesian Information Criterion; $p\ln n - 2\ln\hat{\mathcal{L}}$.

Carrying capacity K
Upper asymptote of a saturation model; identified here with $3^6 = 729$.

Deterrence Strategic situation in which the threat of catastrophic retaliation prevents aggressive action; formalised as $P_{ij} = -\infty$ in the payoff matrix.

Gompertz model
Saturation curve $K\exp(-e^{-r(t-t_0)})$; asymmetric about inflection point.

Hill model Saturation curve $Kt^n/(t_0^n + t^n)$; widely used in pharmacokinetics.

Logistic model Saturation curve $K/(1 + e^{-r(t-t_0)})$; symmetric about inflection t_0.

Lyapunov function
Energy-like function $V \geq 0$ with $\dot{V} \leq 0$; used to prove stability.

oliGARCH Economic agent type in the SNoG framework; one of $3^6 = 729$ sign-vector configurations.

R^2 Coefficient of determination; $1 - \mathrm{SSR}/\mathrm{SST} \in [0, 1]$ (or exactly 1 here).

Recapitalisation
Transfer of wealth w_i per non-oliGARCH in district i subject to $\sum w_i n_i = T$.

Responsibility statistic
$r_i = n_i/o_i$; ratio of non-oliGARCHs to oliGARCHs in district i.

Shapley value Fair cost/surplus allocation in cooperative games [5].

SNoG Standard Nuclear oliGARCHy: 9 districts, 729 oliGARCHs, 48,524 total population, each district nuclear-capable.

SSR Sum of squared residuals; $\sum(P_i - \hat{P}_i)^2$.

SST Total sum of squares; $\sum(P_i - \bar{P})^2$.

Warfare economics
Branch of economics studying resource allocation under strategic conflict, deterrence, and arms-race dynamics [6].

The End